

ML One Network: Norming Variable Power Analysis

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Sample Size Estimation for Subjective Rating Variables

In a follow up to the Semantic Priming Across Many Languages (Buchanan et al. 2025) project, we will collect data across 45 languages/variants on the following norming variables: familiarity, concreteness, valence, arousal, imageability, and age of acquisition (AoA). We will use the Buchanan et al. (2026) accuracy in parameter estimation procedure to determine how many participants we need to create a sufficiently narrow confidence interval around the mean for each item. In this procedure, we use the following steps (reproduced from original article):

Step 1.

Use representative pilot data.

We included all norming datasets found in the original Linguistic Annotated Bibliography (Buchanan et al. 2019) and the updated LAB workflow under construction at <https://manynorms.many-languages.com>. We included datasets with our target variables that included information about the mean and standard deviation of each item. Datasets with only the mean were excluded, as we could not simulate the data appropriately with no information about variance. The original articles were reviewed for information on item sample sizes, which was matched to the variables included in the dataset. In cases of range estimates provided in the manuscript (i.e., 15-20 people rated each item), we used the lower end of the range to estimate with less precision (which would, in turn, increase sample size requirements due to more uncertainty). The following datasets were used:

```
dataset_variables <- read.csv("datasets_with_power_variables.csv", stringsAsFactors = FALSE)

dataset_citation_table <- dataset_variables %>%
  dplyr::mutate(
    bibtex_key = sub("\\.csv$", "", dataset),
    Reference = sprintf("[%s]", bibtex_key),
    Variables = dplyr::case_when(
      TRUE ~ apply(
        dplyr::select(., familiar, concrete, valence, arousal, imagine, aoa), 1,
        function(row) paste(names(row)[row == "x" & !is.na(row)], collapse = ", ")
      )
    ) %>%
  dplyr::transmute(Dataset = bibtex_key, Variables, Reference)

knitr::kable(dataset_citation_table, row.names = FALSE)
```

Dataset	Variables	Reference
Alario1999	familiar, aoa	(Alario and Ferrand 1999)
Alonso2015	aoa	(Alonso et al. 2015)

Dataset	Variables	Reference
Alonso2016	aoa	(Alonso et al. 2016)
Barca2002	familiar, concrete, imagine, aoa	(Barca et al. 2002)
Bennett2011	imagine	(Bennett et al. 2011)
Berrian1979	imagine	(Berrian et al. 1979)
Birchenough2017	aoa	(Birchenough et al. 2017)
Bogunovic2023	familiar, concrete, valence, arousal	(Bogunović 2023)
Bonin2003	familiar, aoa	(Bonin et al. 2003)
Bonin2018	concrete, valence, arousal	(Bonin et al. 2018)
Bonin2021	familiar, imagine, aoa	(Bonin et al. 2021)
Borsa2025	familiar, concrete, valence, arousal, imagine, aoa	(Borsa et al. 2025)
Boukadi2016	familiar, imagine	(Boukadi et al. 2016)
Bradley1999	valence, arousal	(Bradley and Lang 1999)
Brysbaert2014	concrete	(Brysbaert, Warriner, et al. 2014)
Brysbaert2014a	concrete, aoa	(Brysbaert, Stevens, et al. 2014)
Cameirao2010	aoa	(Cameirão and Vicente 2010)
Carroll1973	aoa	(Carroll and White 1973)
Chan2024	familiar, concrete, valence, arousal, imagine	(Chan and Tse 2024)
Cheng2025	valence, arousal	(Cheng et al. 2025)
Citron2016	familiar, concrete, valence, arousal	(Citron et al. 2016)
Clarke2018	familiar, aoa	(Clarke and Ludington 2018)
Clarke2024	imagine, aoa	(Clarke 2024)
Conca2024	familiar, valence, arousal, imagine, aoa	(Conca et al. 2024)
Conde2026_familiar_human	familiar	(Conde et al. 2026)
Cortese2004	imagine	(Cortese and Fugett 2004)
Coso2019	concrete, valence, arousal	(Ćoso et al. 2019)
DeDeyne2008	familiar, imagine	(Deyne and Storms 2008)
DellaRosa2010	familiar, concrete, imagine, aoa	(Rosa et al. 2010)
Dellacqua2000_1	familiar, aoa	(Dell’acqua et al. 2000)
Desrochers2009	imagine	(Desrochers and Thompson 2009)
Desrochers2010	imagine	(Desrochers et al. 2010)
Diez-Alamo2018	aoa	(Díez-Álamo et al. 2019)
Falcinelli2024	familiar, concrete, valence, arousal, imagine, aoa	(Falcinelli et al. 2024)
Ferrand2008	aoa	(Ferrand et al. 2008)
Ferre2012	familiar, concrete, valence, arousal	(Ferré et al. 2012)
Fiez1997	familiar	(Fiez and Tranel 1997)
Friendly1982	concrete	(Friendly et al. 1982)
Gilhooly1977	familiar, concrete, imagine, aoa	(Gilhooly and Hay 1977)
Gilhooly1980	familiar, concrete, imagine, aoa	(Gilhooly and Logie 1980b)
Gilhooly1980a	familiar, concrete, imagine, aoa	(Gilhooly and Logie 1980a)
Granja2025	familiar	(Granja et al. 2025)
Grigoriev2013	familiar, imagine	(Grigoriev and Oshhepkov 2013)
Guasch2016	familiar, concrete, valence, arousal, imagine	(Guasch et al. 2016)
Hermans1994	familiar	(Hermans and Houwer 1994)
Hinojosa2016a	concrete, valence, arousal	(J. A. Hinojosa et al. 2016)

Dataset	Variables	Reference
Hinojosa2016b	familiar	(José A. Hinojosa et al. 2016)
Imbir2015	valence, arousal	(Imbir 2015)
Janschewitz2008	familiar, valence, arousal, imagine	(Janschewitz 2008)
Janssen2011	familiar, imagine	(Janssen et al. 2011)
Kanske2010	concrete, valence, arousal	(Kanske and Kotz 2010)
Kerr1991	familiar, concrete, imagine	(Kerr and Johnson 1991)
Khwaileh2018	familiar, imagine, aoa	(Khwaileh et al. 2018)
Klee1976	concrete	(Klee and Legge 1976)
Kraljevic2018	familiar, concrete, imagine, aoa	(Kraljević and Olujić 2018)
Kuperman2012	aoa	(Kuperman et al. 2012)
Kurdi2017	valence, arousal	(Kurdi et al. 2017)
Lachaud2007	familiar, aoa	(Lachaud 2007)
Li2021	familiar	(Li et al. 2021)
Lippman1974	imagine	(Lippman 1974)
Liu2025	familiar, concrete, valence, arousal	(Liu et al. 2026)
Manoiloff2010	familiar, imagine, aoa	(Manoiloff et al. 2010)
Marques2007	aoa	(Marques et al. 2007)
Masterson1998	familiar, imagine, aoa	(Masterson and Druks 1998)
Miklashevsky2018	imagine, aoa	(Miklashevsky 2018)
Mochizuki2025	aoa	(Mochizuki and Ota 2025)
Monnier2017	valence, arousal	(Monnier and Syssau 2017)
Montefinese2014	familiar, concrete, valence, arousal, imagine	(Montefinese et al. 2014)
Montefinese2019	aoa	(Montefinese 2019)
Montefinese2023_en	concrete	(Montefinese et al. 2023a)
Montefinese2023_it	concrete	(Montefinese et al. 2023b)
Moors2013	valence, arousal, aoa	(Moors et al. 2013)
Moreno-Martinez2012	familiar, aoa	(Moreno-Martínez and Montoro 2012)
Moreno-Martinez2014	familiar	(Moreno-Martínez et al. 2014)
Muraki2025	concrete, valence, arousal, imagine	(Muraki and Pexman 2025)
Navarrete2019	familiar, aoa	(Navarrete et al. 2019)
Paivio1968	concrete, imagine	(Paivio et al. 1968)
PetiStantic2021	concrete, imagine, aoa	(Peti-Stantić et al. 2021)
Quadflieg2014	familiar, concrete, valence	(Quadflieg et al. 2014)
Raman2014	aoa	(Raman et al. 2014)
Rami2022	imagine, aoa	(Rami et al. 2024)
Raslescu2023	valence, imagine	(Raslescu et al. 2023)
Ric2013	valence	(Ric et al. 2013)
Rinaldi2004	imagine	(Rinaldi et al. 2004)
Rocklage2018	valence	(Rocklage et al. 2018)
Roest2017	valence, arousal	(Roest et al. 2018)
Rogic2013	familiar	(Rogić et al. 2013)
Russell1970	familiar, aoa	(Russell 1970)
Ruts2004	aoa	(Ruts et al. 2004)
Salmon2010	familiar	(Salmon et al. 2010)
Schauenburg2015	valence, arousal	(Schauenburg et al. 2015)
Schmidtke2014	valence, imagine	(Schmidtke et al. 2014)
Schock2012	aoa	(Schock et al. 2012)
Schroder2012	familiar, aoa	(Schröder et al. 2012)

Dataset	Variables	Reference
Schwitter2004	familiar, aoa	(Schwitter et al. 2004)
Scott2017	familiar, concrete, valence, arousal, imagine, aoa	(Scott et al. 2019a)
Scott2019	familiar, concrete, valence, arousal, imagine, aoa	(Scott et al. 2019b)
Siew2025	concrete, valence, arousal	(Siew et al. 2025)
Snodgrass1980	familiar	(Snodgrass and Vanderwart 1980)
Soares2012	valence, arousal	(Soares et al. 2012)
Soares2017	concrete, imagine	(Soares et al. 2017)
Stadthagen-Gonzalez2017	valence, arousal	(Stadthagen-Gonzalez et al. 2017)
Stadthagen-Gonzalez2017a	valence, arousal	(Stadthagen-González et al. 2018)
Stratton1975	familiar, imagine, aoa	(Stratton et al. 1975)
Surprenant1999	familiar	(Surprenant et al. 1999)
Syssau2009	valence	(Syssau and Monnier 2009)
Tabossi2011	familiar, aoa	(Tabossi et al. 2011)
Toglia1978	familiar, concrete, imagine	(Toglia 2009)
Tsaparina2011	familiar, imagine, aoa	(Tsaparina et al. 2011)
Valueva2025	aoa	(Valueva and Grigoriev 2025)
VanderGoten1999	concrete, valence	(Goten et al. 1999)
Verheyen2020	familiar, concrete, valence, arousal, imagine, aoa	(Verheyen et al. 2020)
Vo2009	valence, arousal, imagine	(Võ et al. 2009)
Walker1970	imagine	(Walker 1970)
Walley1992	imagine	(Walley and Metsala 1992)
Warriner2013	valence, arousal	(Warriner et al. 2013)
Whalen1994	familiar	(Whalen and Zsiga 1994)
Xu2021	aoa	(Xu et al. 2021)
Yao2017	familiar, concrete, valence, arousal, imagine	(Yao et al. 2017)
Yee2017	familiar, concrete, valence, arousal, imagine	(Yee 2017)
Yoon2004	familiar	(Yoon et al. 2004)

Step 2.

Calculate the standard error of each of the items in the pilot data. Using the 4th decile, determine the cutoff and stopping rule for the standard error of the items.

Using the *semanticprimeR* package (Buchanan 2024), we then simulated the pilot data using the mean, standard deviation, and sample size from each dataset for the items included. This dataset was treated as the “population”, and we estimated the 4th decile standard error across all items as the cutoff score and stopping rule for the standard error of the items.

Step 3.

Create simulated samples of your pilot data, starting with at least 20 participants, up to a maximum number of participants.

We then sampled from the simulated pilot data, from $n = 20$ up to $n = 500$. However, running the simulation over many samples by many simulations by every dataset would be extremely slow. Therefore, we broke the simulations into two stages:

1. Pilot - a coarse sweep per dataset (sample sizes 20-500, step 10, 10 simulations per step) to find roughly where the precision curve crosses 70% and 95% rates. `run_power_pilot.R`

2. Refine - a narrow, high-precision rerun (step 5, 100 simulations per step) restricted to the pilot's recommended `[start, stop]` window for that specific dataset. `run_power_refine.R`

Both of these scripts call `run_power_batch.R` to do the simulations. This document reads the refined stage output only, since that is the higher-precision, final estimate.

Step 4.

Calculate the standard error of each of the items in the simulated data. From these scores, calculate the percent of items below the cutoff score from Step 2.

Step 5.

Determine the sample size at which 80%, 85%, 90%, 95% of items are below the cutoff score. Use the correction formula to adjust your proposed sample size based on pilot data size, power, and percent variability.

`functions.R` implements steps 3-5 using tools from the `semanticprimerR` package for each real dataset in `power_job_list.csv`.

Step 6.

Report all values. Designate one as the minimum sample size, the cutoff score as the stopping rule for adaptive designs, and the maximum sample size.

Split-half Reliability

We additionally included a calculation of split-half reliability for each of the sample sizes. Participants for each item are split in half, the two half-means are correlated, and the Spearman-Brown correction is then applied. This result tells us how consistent the item-level scores would be with a given number of raters per item. Generally, numbers can be very low and obtain reliability. We tested reliability at sample sizes of $n = 5$ to the stop of the range found for each dataset from the pilot sample size estimation steps.

Target Criteria and Outlier Handling

We target 80% power (AIPE precision) and 80% split-half reliability, and report the recommended sample size per variable as the average of those two numbers. Precision and reliability answer different questions about the same design (how accurately measured is the estimated mean vs. how consistent are the item scores), so averaging them gives one practical number rather than always taking the stricter of the two. We tested $n = 5$, but set the bottom value for reliability to be $n = 20$. Therefore, that means even if samples achieved high reliability at $n = 5, 10$, or 15 , the lowest number we would accept for the average of criterions would be $n = 20$, as otherwise the low numbers might bias the sample size estimates downward inappropriately.

Each of these numbers is computed per dataset run (i.e., per source paper/corpus) before pooling: for a given variable, every dataset run contributes one “n needed for 80% precision” value and one “n needed for 80% reliability” value. A handful of source datasets are small, noisy, or otherwise unusual and can pull these per-run numbers to extreme highs or lows. Before pooling across runs, we drop outlier runs using the standard $1.5 \times \text{IQR}$ rule (values more than 1.5 interquartile ranges below $Q1$ or above $Q3$ are dropped), applied separately to the precision values and the reliability values for that variable. We then take the median of the remaining runs for each metric (median rather than mean, since a handful of runs can still sit near the outlier boundary without being dropped outright - in practice the two are close once outliers are trimmed, but median is the more standard choice for a “typical” recommendation). The per-variable sections below show which runs were considered outliers.

Loading Refined Results

```

variable_dirs <- file.path("simulations", VARIABLES, "refined")
names(variable_dirs) <- VARIABLES
available_vars <- VARIABLES[dir.exists(variable_dirs) &
  lengths(purrr::map(variable_dirs, list.files, pattern = "\\.*rds$")) > 0]

cat("Variables with refined output available:", paste(available_vars, collapse = ", "), "\n")

## Variables with refined output available: familiar, concrete, valence, arousal, imagine, aoa
cat("Variables still pending:", paste(setdiff(VARIABLES, available_vars), collapse = ", "), "\n")

## Variables still pending:
results_by_variable <- purrr::map(
  setNames(available_vars, available_vars),
  ~ load_and_summarize_simulations(file.path("simulations", .x, "refined"))
)

```

Each dataset contributes one simulation “run” per variable (i.e., one row per source paper/corpus that reported that variable for the words in our stimuli set). The tables below pool across *all* dataset runs for a variable, since our own norming study will draw one sample of participants per item. Each of these datasets is a mix of languages, so we estimate our overall sample size globally, rather than by language.

Per-Variable Results

```

target_results_by_variable <- purrr::map(results_by_variable, ~ summarize_by_target(.x$overall_power))
labeled_power_by_variable <- purrr::map(results_by_variable, ~ label_by_target(.x$overall_power))
variable_recommendation <- purrr::map(results_by_variable, ~ summarize_variable_recommendation(.x))

variable_labels <- c(
  familiar = "Familiarity", concrete = "Concreteness", valence = "Valence",
  arousal = "Arousal", imagine = "Imageability", aoa = "Age of Acquisition (AoA)"
)

for (v in available_vars) {
  cat(sprintf("\n### %s\n\n", variable_labels[[v]]))

  res <- results_by_variable[[v]]
  target_summary <- target_results_by_variable[[v]]
  n_runs <- dplyr::n_distinct(res$overall_power$run_id)
  cat(sprintf("Based on %d dataset run(s) for `%s`.\n\n", n_runs, v))

  cat("**Recommended sample size per item, by target precision (% of simulations hitting the AIPE cutoff")
  print(htmltools::tagList(DT::datatable(
    target_summary %>%
      dplyr::mutate(dplyr::across(where(is.numeric), ~ round(.x, 2))),
    rownames = FALSE, options = list(dom = "t", pageLength = 10)
  )))

  box_dat <- labeled_power_by_variable[[v]] %>%
    dplyr::mutate(target_power = factor(target_power, levels = REFINE_POWER_LEVELS))
  print(
    ggplot2::ggplot(box_dat, ggplot2::aes(x = target_power, y = corrected_sample_size)) +
    ggplot2::geom_boxplot(fill = "#d9e7ff", outlier.alpha = 0.2, width = 0.6) +
    ggplot2::geom_jitter(width = 0.12, alpha = 0.35, size = 1.4) +
  
```

```

ggplot2::geom_line(
  data = target_summary %>% dplyr::mutate(target_power = factor(target_power, levels = REFINE_POWER),
  ggplot2::aes(x = target_power, y = recommended_sample_size, group = 1),
  color = "#b00020", linewidth = 0.9
) +
ggplot2::geom_point(
  data = target_summary %>% dplyr::mutate(target_power = factor(target_power, levels = REFINE_POWER),
  ggplot2::aes(x = target_power, y = recommended_sample_size),
  color = "#b00020", size = 2.5
) +
ggplot2::labs(
  x = "Target precision (%)", y = "Corrected sample size",
  title = paste0(v, ": spread of per-dataset recommendations"),
  subtitle = "red line = across-dataset running max of the median"
) +
ggplot2::theme_minimal()
)

if (nrow(res$reliability_summary) > 0) {
  cat("\n\n**Sample size needed to reach target split-half reliability:**\n\n")
  print(htmltools::tagList(DT::datatable(
    res$reliability_recommendation %>%
      dplyr::mutate(dplyr::across(where(is.numeric), ~ round(.x, 3))),
    rownames = FALSE, options = list(dom = "t", pageLength = 10)
  )))

  if (!is.null(res$reliability_plot)) {
    print(res$reliability_plot +
      ggplot2::labs(subtitle = paste0(v, ": median reliability curve across dataset runs")))
  }
}

rec <- variable_recommendation[[v]]
cat(sprintf(
  "\n\n**Recommended n (80%% precision + 80%% reliability, IQR-trimmed medians averaged): %.1f -> %d\n\n",
  rec$recommended_n, ceiling(rec$recommended_n)
))
cat(sprintf(
  "- Precision: %.1f (median of %d/%d runs kept after IQR trim)\n",
  rec$precision_median, rec$precision_n_kept, rec$precision_n_total
))
cat(sprintf(
  "- Reliability: %.1f (median of %d/%d runs kept after IQR trim)\n",
  rec$reliability_median, rec$reliability_n_kept, rec$reliability_n_total
))

cat("\n\n**Per-dataset detail (n at 80% precision, n at 80% reliability, outlier flags):**\n\n")
print(htmltools::tagList(DT::datatable(
  rec$per_run %>%
    dplyr::mutate(dplyr::across(c(n_precision, n_reliability), ~ round(.x, 1))) %>%
    dplyr::arrange(dplyr::desc(n_precision)),
  rownames = FALSE, options = list(pageLength = 10)
)))

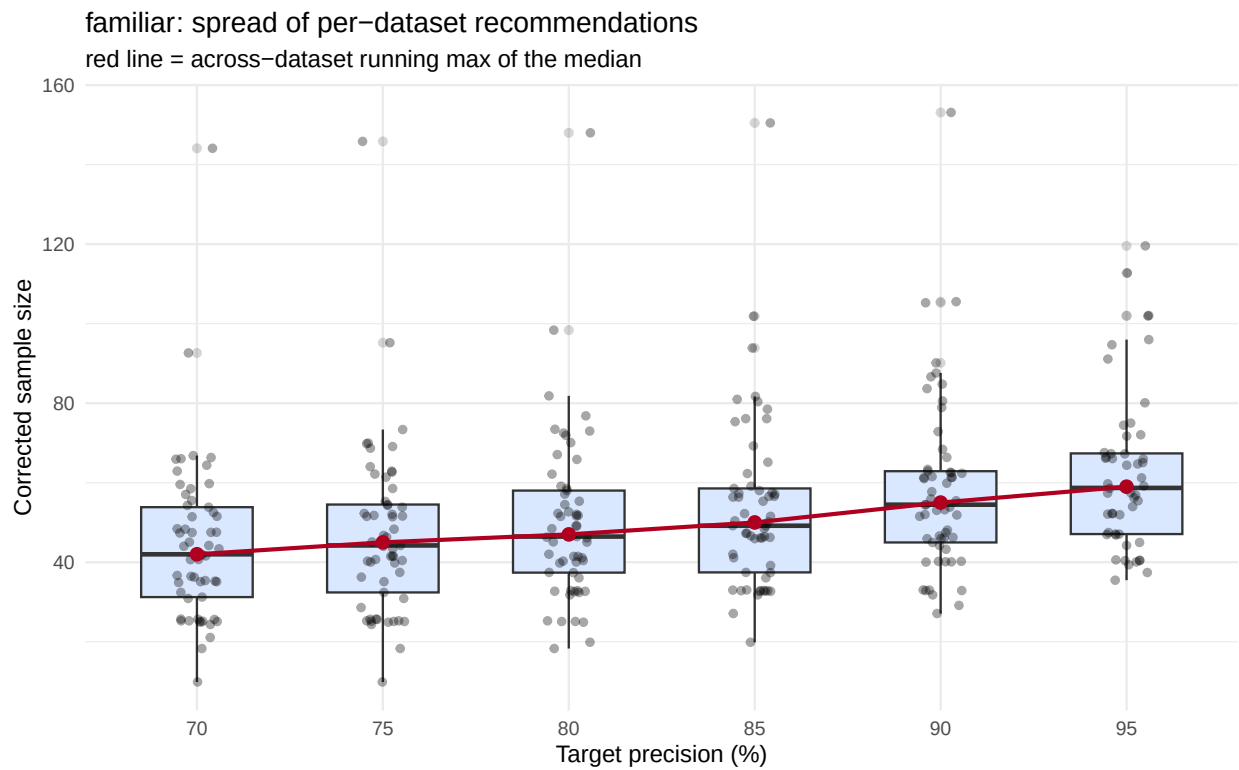
```

```
cat("\n\n---\n")
}
```

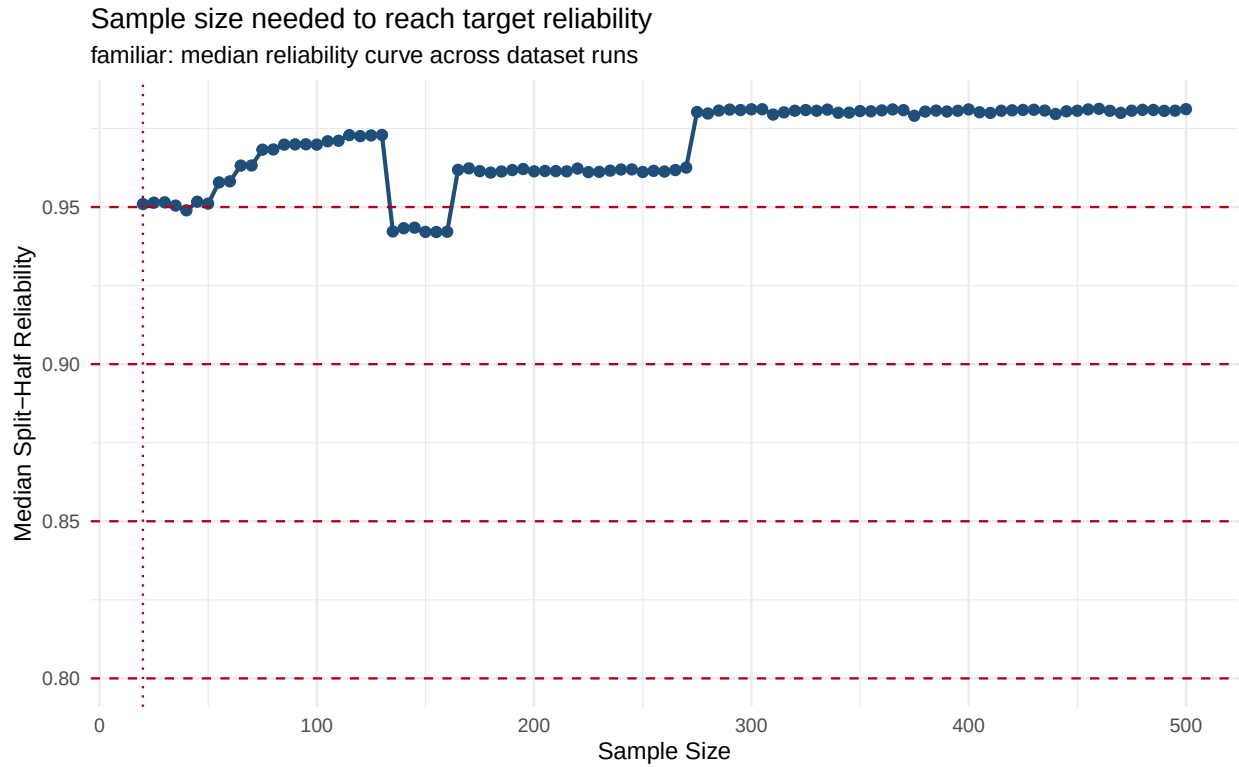
Familiarity

Based on 57 dataset run(s) for `familiar`.

Recommended sample size per item, by target precision (% of simulations hitting the AIPE cutoff):



Sample size needed to reach target split-half reliability:



Recommended n (80% precision + 80% reliability, IQR-trimmed medians averaged): 33.2 -> 34 participants/item

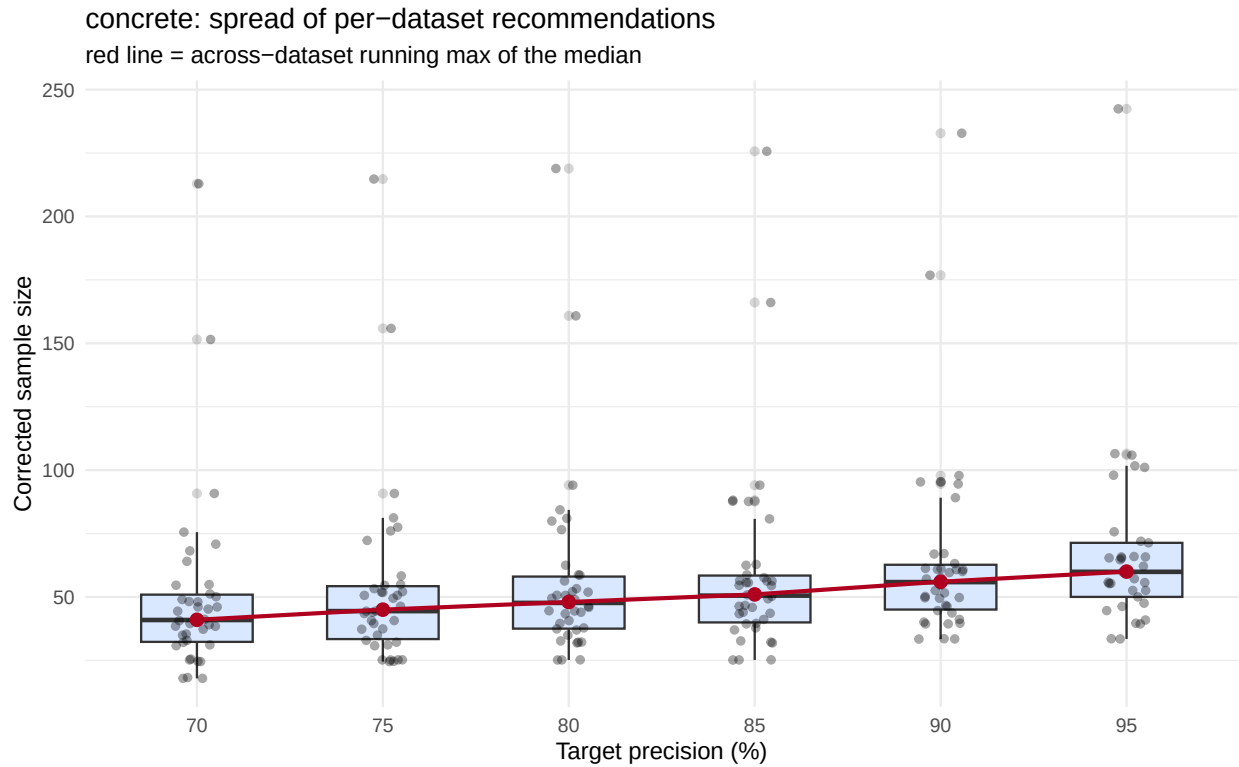
- Precision: 46.3 (median of 55/57 runs kept after IQR trim)
- Reliability: 20.0 (median of 56/57 runs kept after IQR trim)

Per-dataset detail (n at 80% precision, n at 80% reliability, outlier flags):

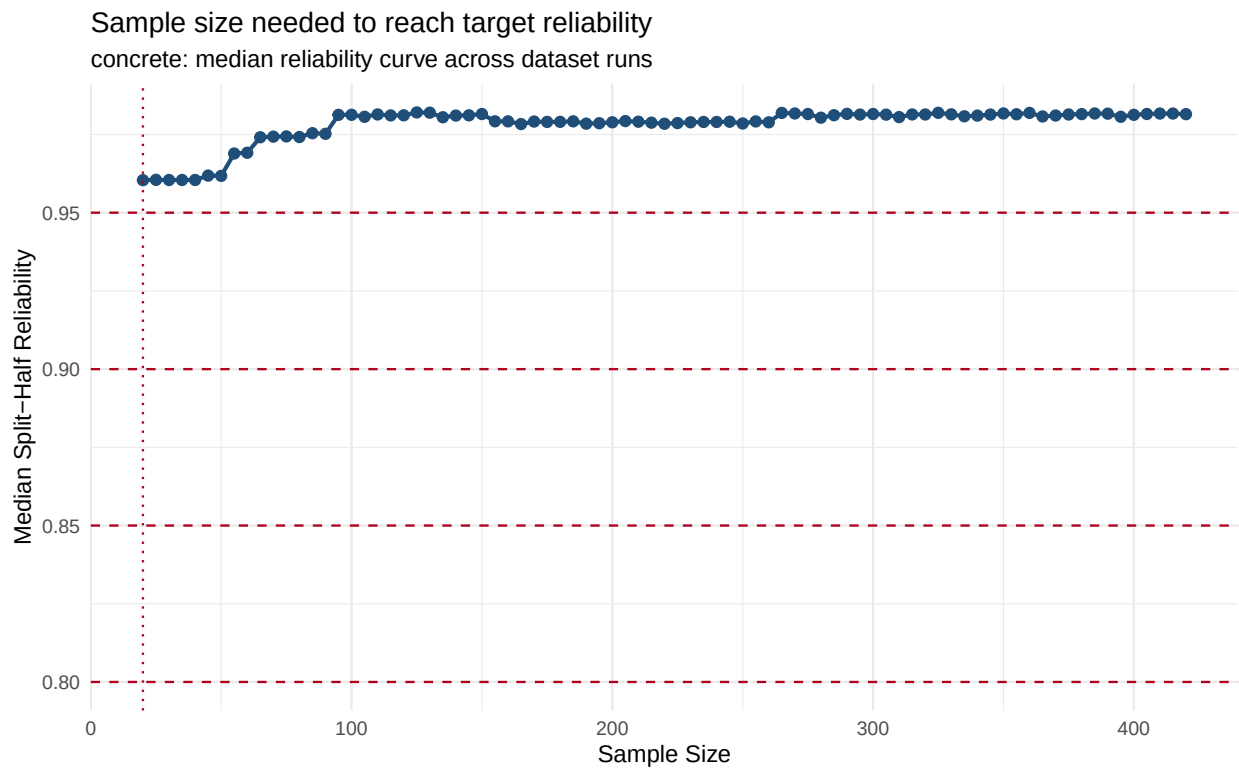
Concreteness

Based on 38 dataset run(s) for concrete.

Recommended sample size per item, by target precision (% of simulations hitting the AIPE cutoff):



Sample size needed to reach target split-half reliability:



Recommended n (80% precision + 80% reliability, IQR-trimmed medians averaged): 32.9 -> 33 participants/item

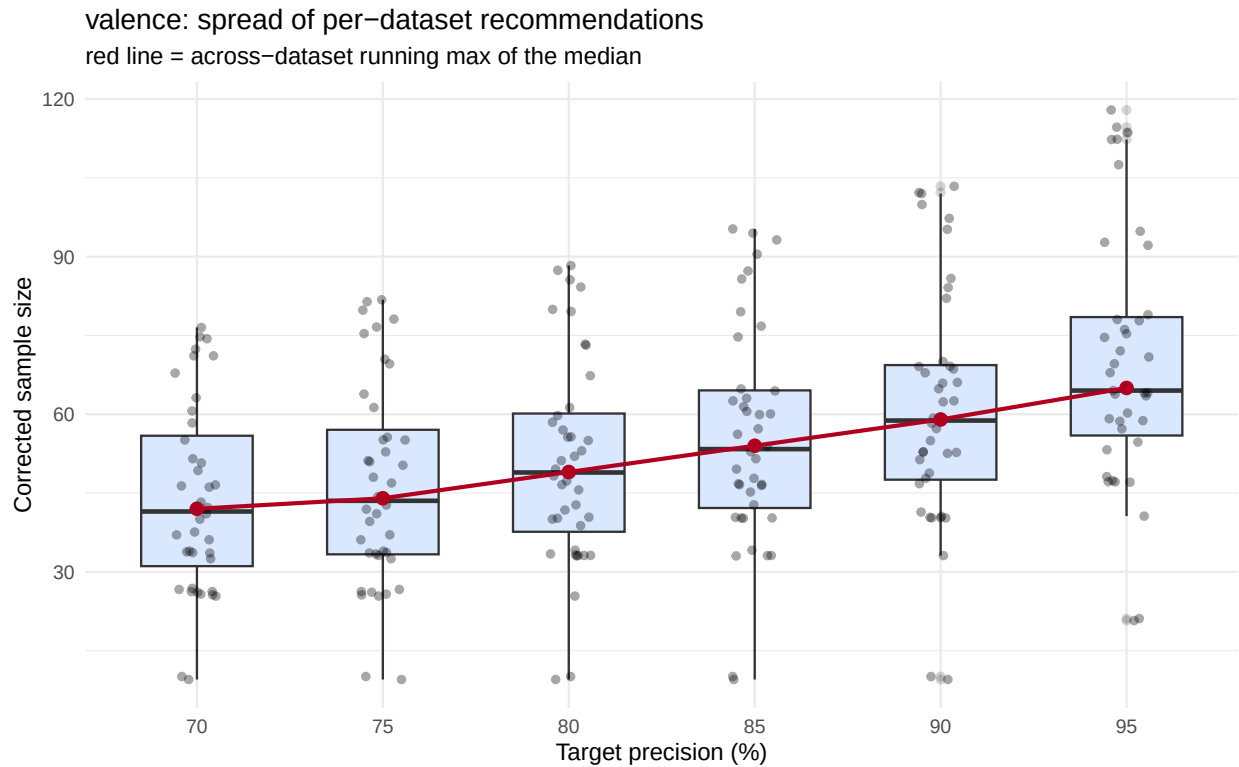
- Precision: 45.9 (median of 35/38 runs kept after IQR trim)
- Reliability: 20.0 (median of 38/38 runs kept after IQR trim)

Per-dataset detail (n at 80% precision, n at 80% reliability, outlier flags):

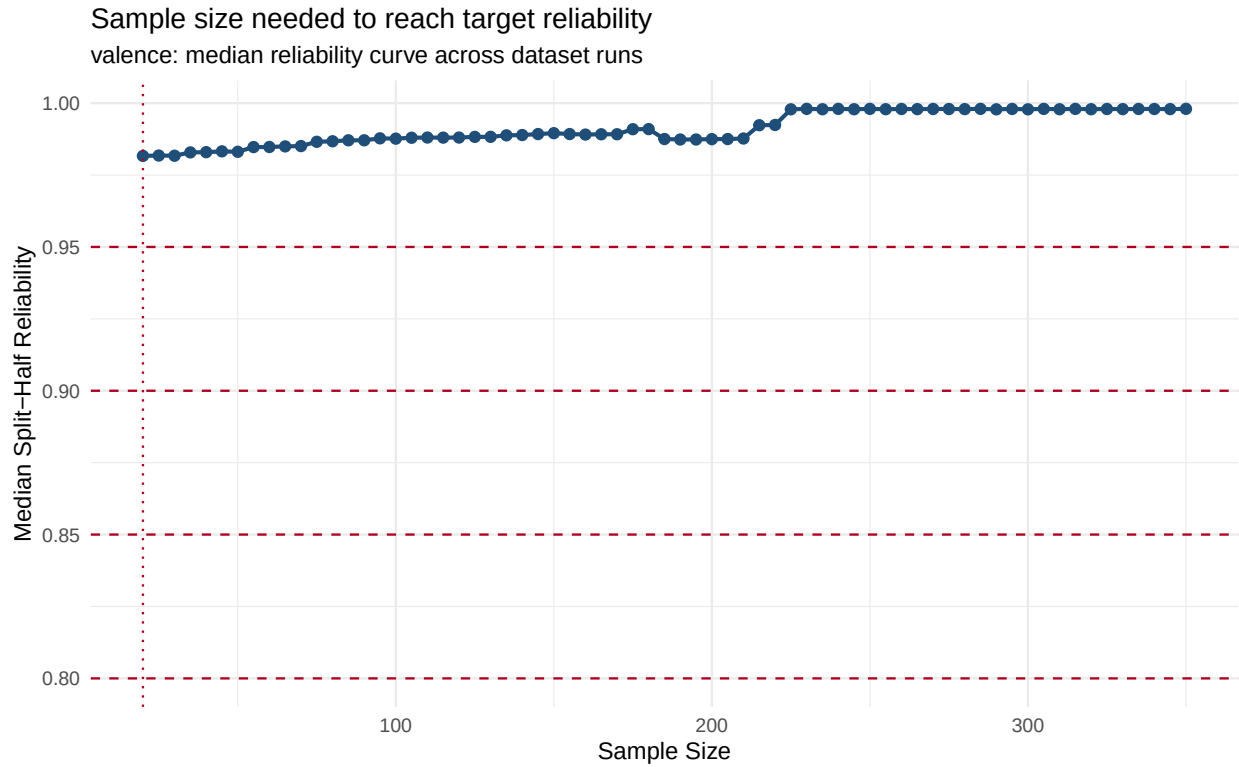
Valence

Based on 40 dataset run(s) for valence.

Recommended sample size per item, by target precision (% of simulations hitting the AIPE cutoff):



Sample size needed to reach target split-half reliability:



Recommended n (80% precision + 80% reliability, IQR-trimmed medians averaged): 34.5 -> 35 participants/item

- Precision: 48.9 (median of 40/40 runs kept after IQR trim)
- Reliability: 20.0 (median of 40/40 runs kept after IQR trim)

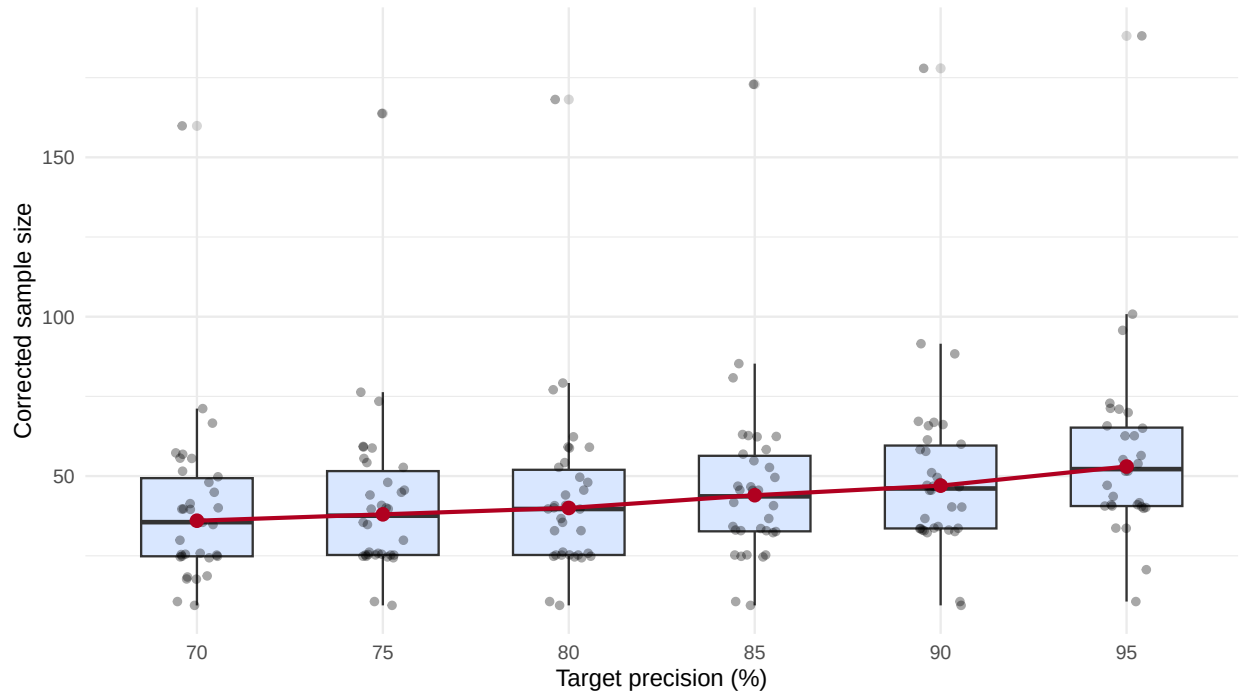
Per-dataset detail (n at 80% precision, n at 80% reliability, outlier flags):

Arousal

Based on 34 dataset run(s) for arousal.

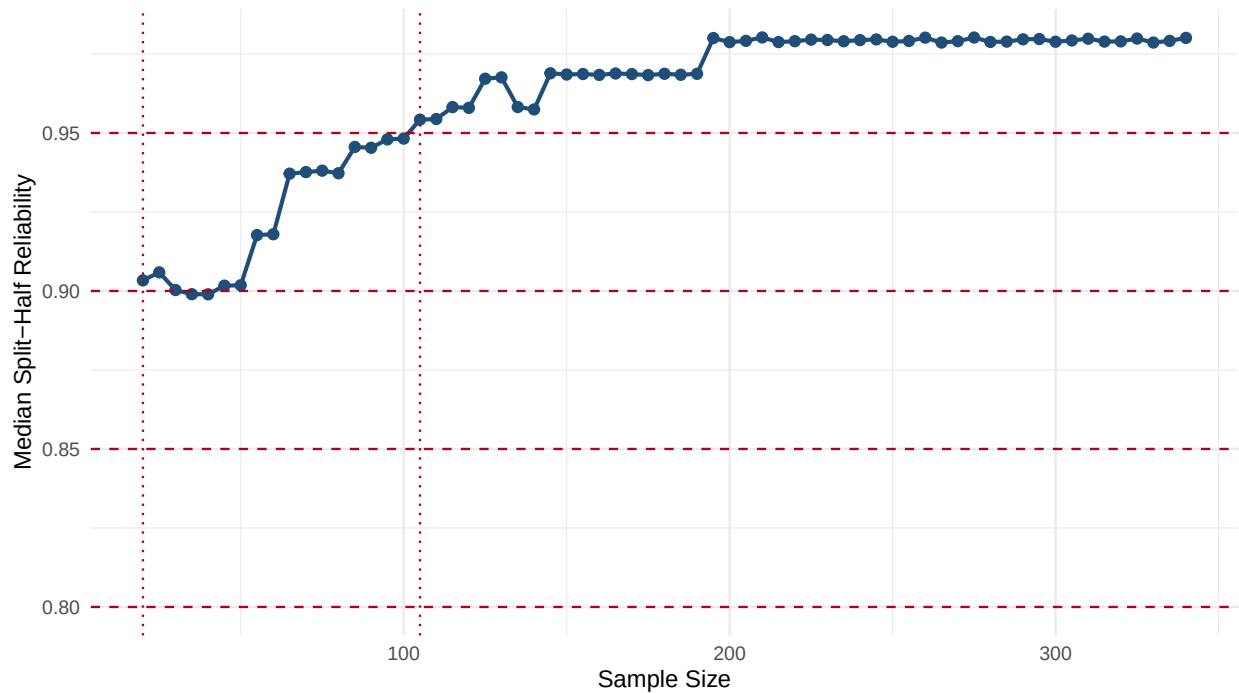
Recommended sample size per item, by target precision (% of simulations hitting the AIPE cutoff):

arousal: spread of per-dataset recommendations
 red line = across-dataset running max of the median



Sample size needed to reach target split-half reliability:

Sample size needed to reach target reliability
 arousal: median reliability curve across dataset runs



Recommended n (80% precision + 80% reliability, IQR-trimmed medians averaged): 29.8 -> 30 participants/item

- Precision: 39.7 (median of 33/34 runs kept after IQR trim)
- Reliability: 20.0 (median of 29/29 runs kept after IQR trim)

Per-dataset detail (n at 80% precision, n at 80% reliability, outlier flags):

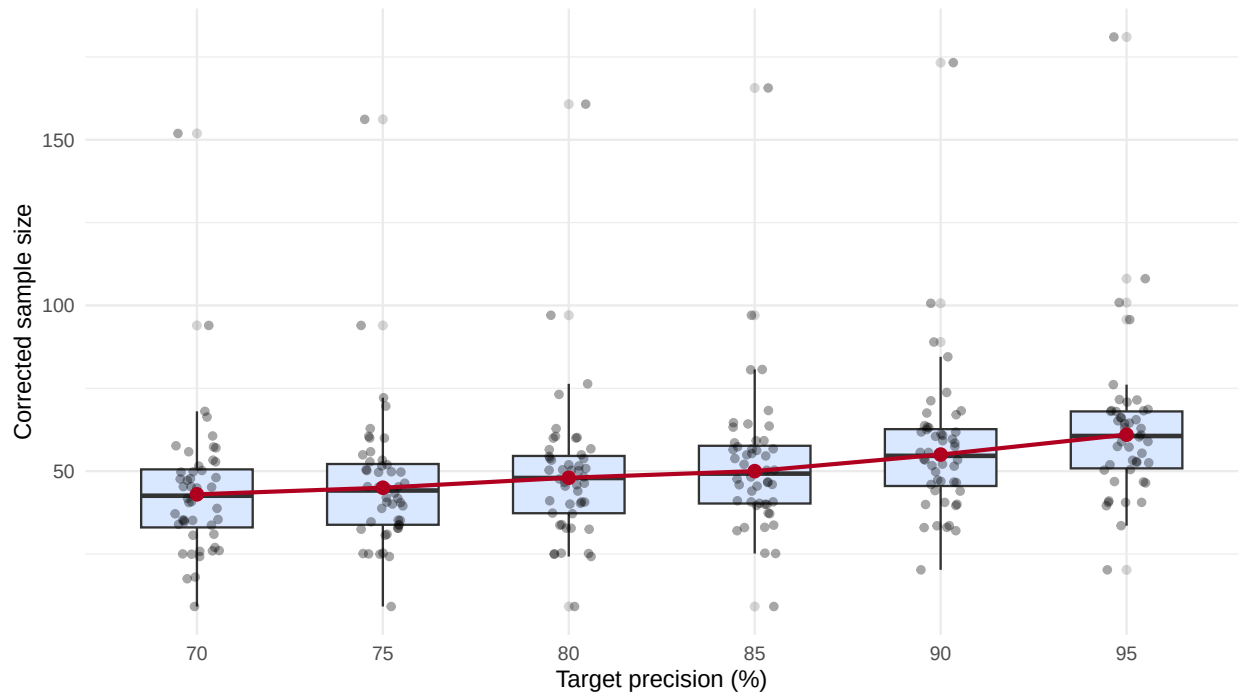
Imageability

Based on 48 dataset run(s) for imagine.

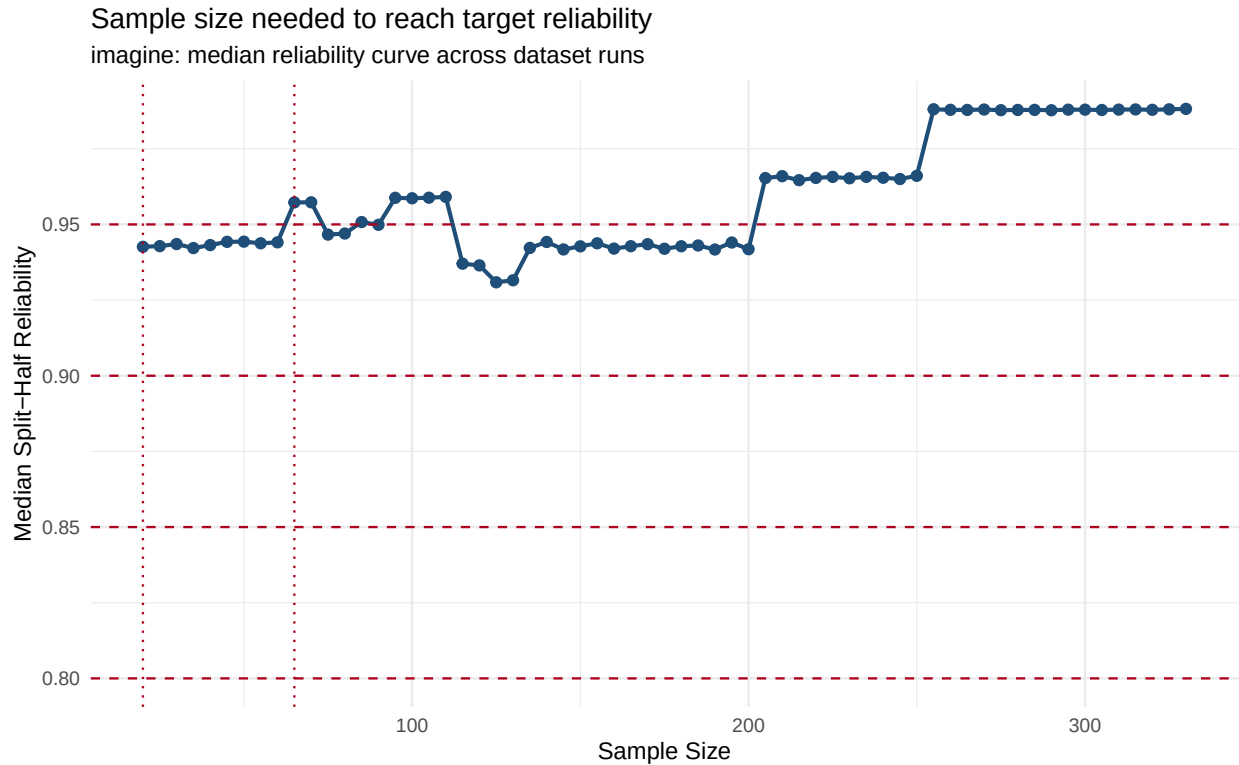
Recommended sample size per item, by target precision (% of simulations hitting the AIPE cutoff):

imagine: spread of per-dataset recommendations

red line = across-dataset running max of the median



Sample size needed to reach target split-half reliability:



Recommended n (80% precision + 80% reliability, IQR-trimmed medians averaged): 33.8 -> 34 participants/item

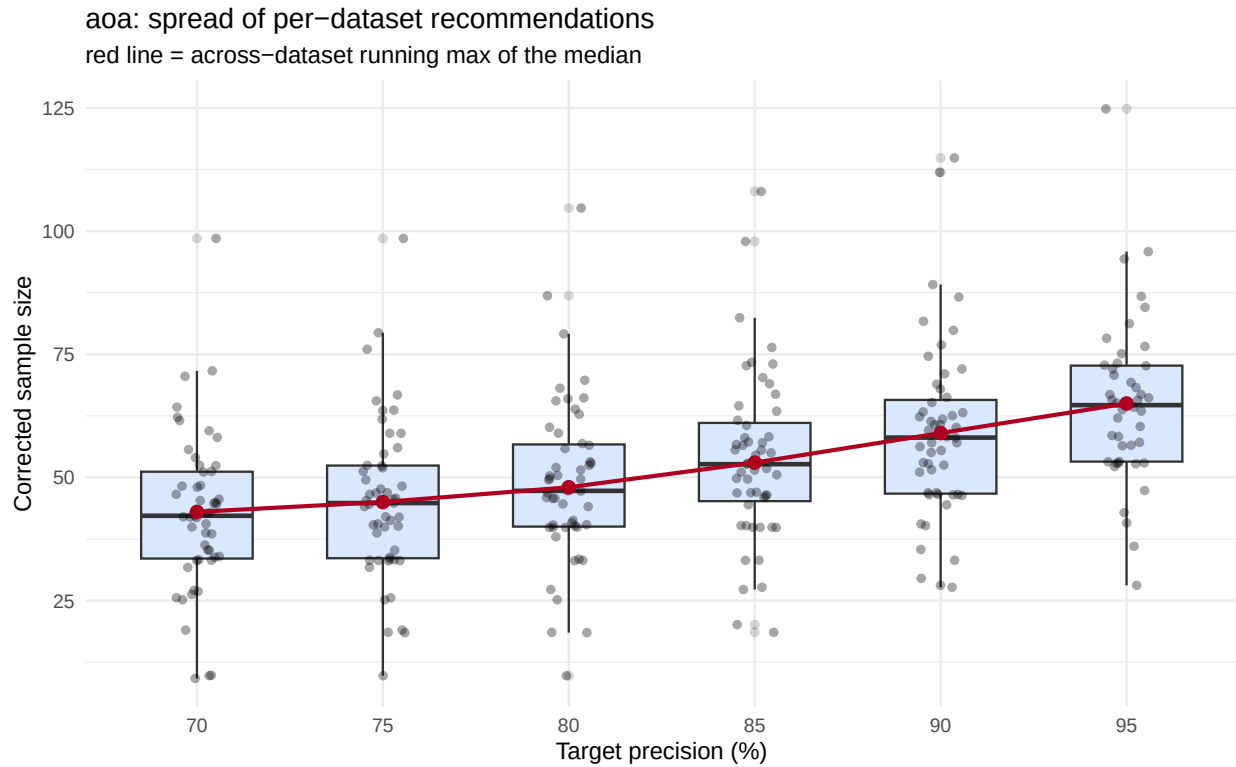
- Precision: 47.6 (median of 45/48 runs kept after IQR trim)
- Reliability: 20.0 (median of 47/47 runs kept after IQR trim)

Per-dataset detail (n at 80% precision, n at 80% reliability, outlier flags):

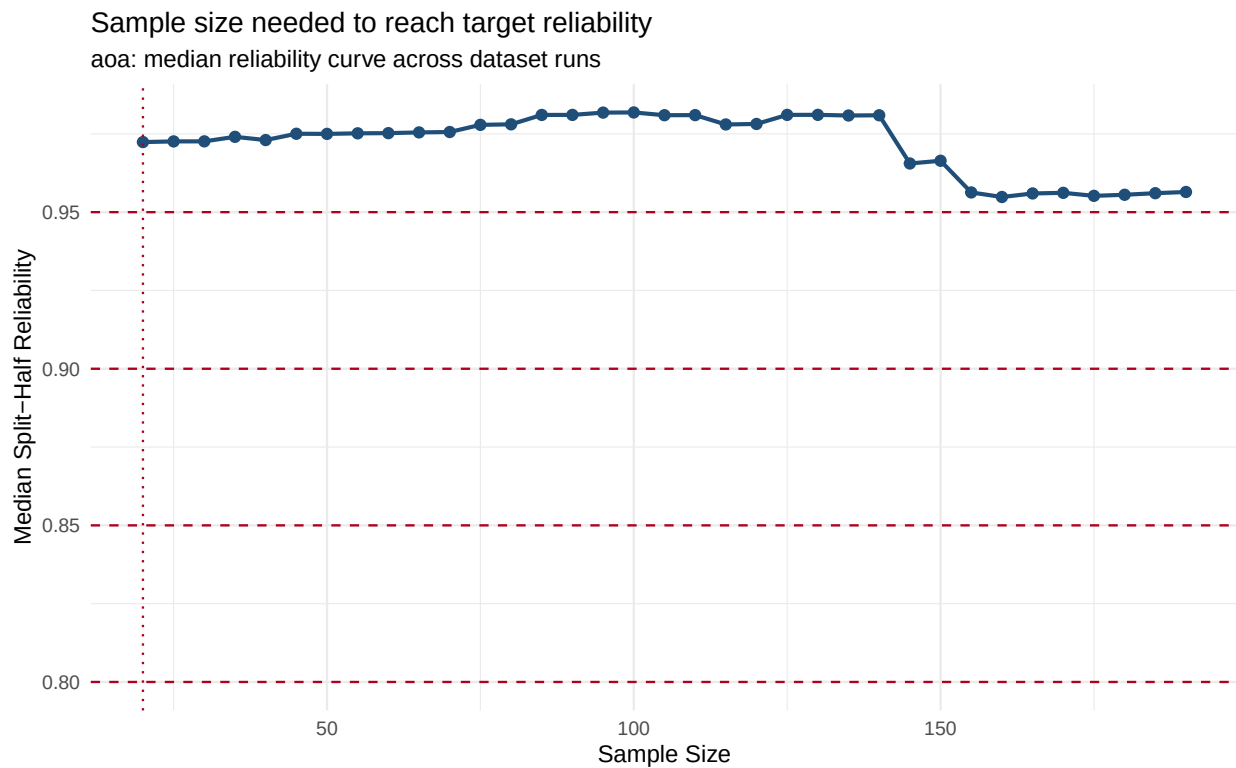
Age of Acquisition (AoA)

Based on 51 dataset run(s) for aoa.

Recommended sample size per item, by target precision (% of simulations hitting the AIPE cutoff):



Sample size needed to reach target split-half reliability:



Recommended n (80% precision + 80% reliability, IQR-trimmed medians averaged): 33.6 -> 34 participants/item

- Precision: 47.3 (median of 48/51 runs kept after IQR trim)
- Reliability: 20.0 (median of 50/50 runs kept after IQR trim)

Per-dataset detail (n at 80% precision, n at 80% reliability, outlier flags):

Combined Recommendation Across Variables

Our norming study collects all six variables from the same participants per item, so the design has to satisfy every variable's IQR-trimmed 80/80 recommendation at once - the number is the largest `recommended_n` across variables below.

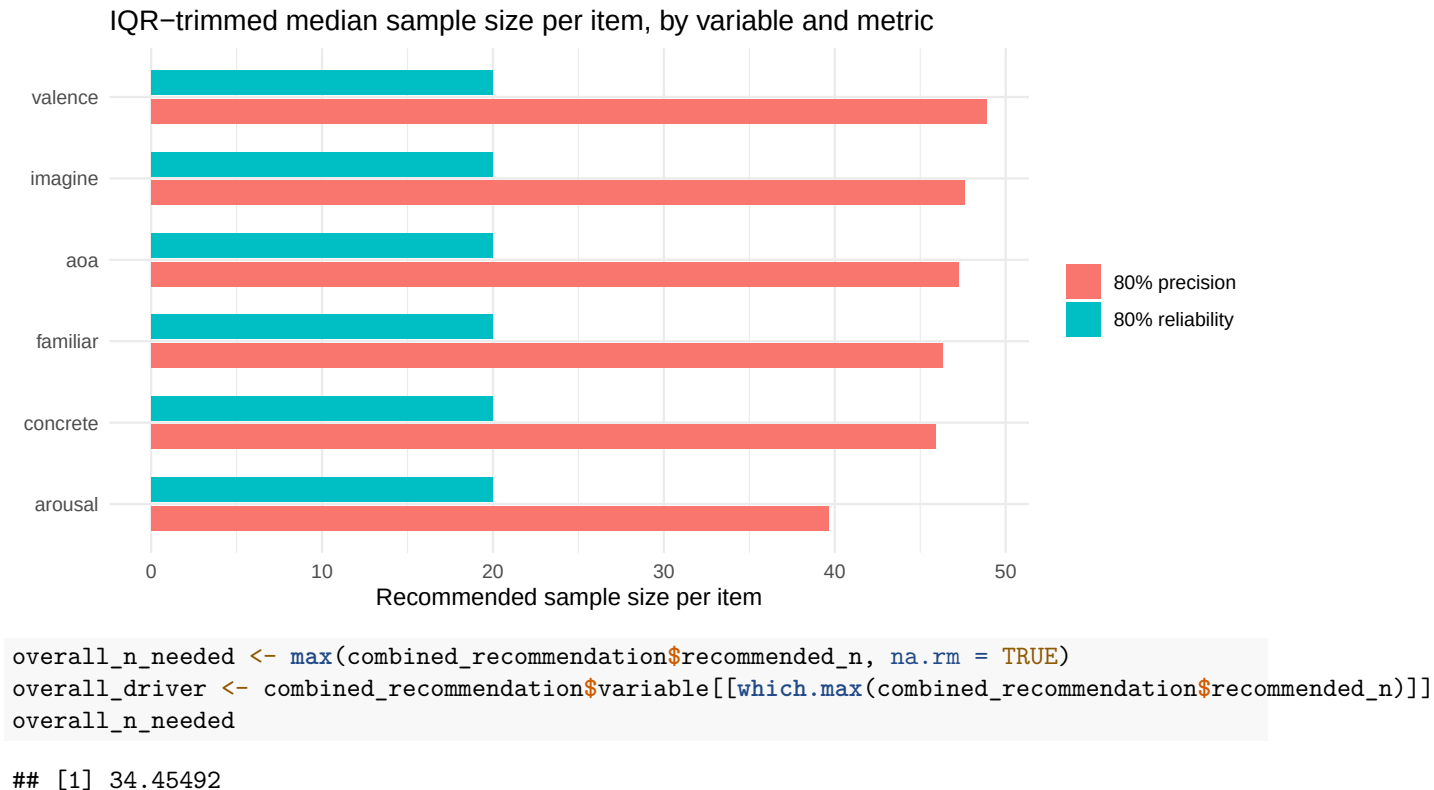
```
combined_recommendation <- purrr::imap_dfr(variable_recommendation, function(rec, v) {
  tibble::tibble(
    variable = v,
    n_precision = rec$precision_median,
    precision_kept = sprintf("%d/%d", rec$precision_n_kept, rec$precision_n_total),
    n_reliability = rec$reliability_median,
    reliability_kept = sprintf("%d/%d", rec$reliability_n_kept, rec$reliability_n_total),
    recommended_n = rec$recommended_n
  )
})

combined_recommendation_display <- combined_recommendation %>%
  dplyr::mutate(dplyr::across(c(n_precision, n_reliability, recommended_n), ~ round(.x, 1))) %>%
  dplyr::arrange(dplyr::desc(recommended_n))

DT::datatable(combined_recommendation_display, rownames = FALSE, options = list(dom = "t", pageLength = 10))

plot_dat <- combined_recommendation %>%
  dplyr::select(variable, n_precision, n_reliability) %>%
  tidyr::pivot_longer(c(n_precision, n_reliability), names_to = "metric", values_to = "n") %>%
  dplyr::mutate(metric = dplyr::recode(metric, n_precision = "80% precision", n_reliability = "80% reliability"))

ggplot2::ggplot(plot_dat, ggplot2::aes(x = reorder(variable, n), y = n, fill = metric)) +
  ggplot2::geom_col(position = ggplot2::position_dodge(width = 0.7), width = 0.6) +
  ggplot2::coord_flip() +
  ggplot2::labs(
    x = NULL, y = "Recommended sample size per item",
    fill = NULL,
    title = "IQR-trimmed median sample size per item, by variable and metric"
  ) +
  ggplot2::theme_minimal()
```



Bottom line: averaging 80% precision and 80% reliability (after dropping outlier dataset runs via the 1.5 x IQR rule) puts every variable's recommended sample size at roughly 30-35 participants per item, with valence needing the most. Our norming study needs at least 35 participants per item to satisfy every variable at once.

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